Math 241, F04 **Name:**

Quiz 8 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (10 points) Evaluate the following integrals.

a) $\int_0^3 \int_0^1 ye^{xy}\, dxdy$

Solution:

$$\begin{aligned}
\int_0^3 \int_0^1 ye^{xy}\, dxdy &= \int_0^3 \left[e^{xy}\right]_0^1 dy \\
&= \int_0^3 e^y - 1\, dy \\
&= \left[e^y - y\right]_0^3 \\
&= (e^3 - 3) - (1 - 0) = e^3 - 4
\end{aligned}$$

b) $\int_0^2 \int_{x-1}^{x+1} x + y\, dydx$

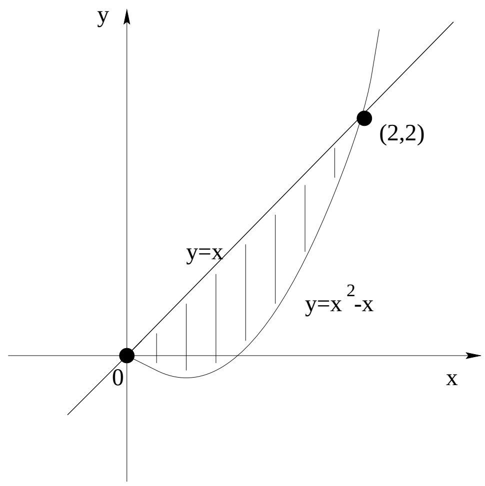
Solution:

$$\begin{aligned}
\int_0^2 \int_{x-1}^{x+1} x + y\, dydx &= \int_0^2 \left[xy + \frac{y^2}{2}\right]_{x-1}^{x+1} dx \\
&= \int_0^2 \left[x(x+1) + \frac{(x+1)^2}{2}\right] - \left[x(x-1) + \frac{(x-1)^2}{2}\right] dx \\
&= \int_0^2 \left[x^2 + x + \frac{x^2 + 2x + 1}{2}\right] - \left[x^2 - x + \frac{x^2 - 2x + 1}{2}\right] dx \\
&= \int_0^2 4x\, dx \\
&= \left[2x^2\right]_0^2 = 8
\end{aligned}$$

Problem 2. (10 points)

a) Let S be the region between $y = x^2 - x$ and $y = x$. Sketch it and write it down in the form of y-simple set.

Solution:



$$S = \{(x, y) \mid 0 \le x \le 2,\ x^2 - x \le y \le x\}$$

b) Evaluate $\displaystyle\iint_S (x^2 - xy)\, dA$.

Solution:

$$\begin{aligned}
\iint_S (x^2 - xy)\, dA &= \int_0^2 \int_{x^2-x}^{x} x^2 - xy\ dydx \\
&= \int_0^2 \left[x^2 y - \frac{xy^2}{2}\right]_{x^2-x}^{x} dx \\
&= \int_0^2 \left[x^3 - \frac{x^3}{2}\right] - \left[x^2(x^2 - x) - \frac{x(x^2-x)^2}{2}\right] dx \\
&= \int_0^2 \frac{x^5}{2} - 2x^4 + 2x^3\ dx \\
&= \left[\frac{x^6}{12} - \frac{2x^5}{5} + \frac{x^4}{2}\right]_0^2 \\
&= \frac{64}{12} - \frac{64}{5} + 8 = \frac{8}{15}
\end{aligned}$$